

FERTILIZER & NUTRIENT MANAGEMENT

RAG Knowledge Base for Agricultural Assistant • Version 1.0 • 2026

Purpose: Provide a retrieval-ready knowledge base for farmer questions about fertilizers, plant nutrients, nutrient deficiencies, application stages, and safe nutrient management. The document is designed to work together with Crop Guides, Disease & Pest Management, Agriculture Guidelines, and Government Schemes.

1. CORE NUTRIENT MANAGEMENT PRINCIPLES

- Use soil-test-based recommendations whenever available. Fertilizer type and rate should match soil nutrient status and crop requirement.
- Follow the 4R principle: Right Source, Right Rate, Right Time, and Right Place.
- Use Integrated Nutrient Management (INM): combine appropriate mineral fertilizers with organic manures, crop residues, green manures and suitable biofertilizers.
- Do not assume that yellow leaves always mean nitrogen deficiency. Water stress, root problems, pH, salinity, pests, diseases and several nutrient deficiencies can produce similar symptoms.
- Avoid indiscriminate or excessive use of urea/DAP. Excess nutrients can reduce nutrient-use efficiency and harm soil and environmental health.
- For an exact fertilizer dose, collect crop, variety, season, location, soil-test status, irrigation method, previous crop and crop-stage information where possible.

2. PLANT NUTRIENTS

Group	Nutrients	Main role / RAG cues
Primary macronutrients	Nitrogen (N)	Leaf and vegetative growth; protein formation; chlorophyll.
Primary macronutrients	Phosphorus (P)	Root development, energy transfer, early establishment and reproductive development.
Primary macronutrients	Potassium (K)	Water regulation, enzyme activity, stress tolerance and crop quality.
Secondary nutrients	Calcium (Ca)	Cell walls, growing points and structural development.
Secondary nutrients	Magnesium (Mg)	Central component of chlorophyll; photosynthesis.
Secondary nutrients	Sulphur (S)	Protein/amino-acid synthesis and enzyme functions.
Micronutrients	Zn, Fe, B, Mn, Cu, Mo, Cl, Ni	Required in small quantities but essential for specific metabolic functions.

3. COMMON FERTILIZER & NUTRIENT SOURCES

Input	Typical nutrient/use category	RAG note
Urea	Nitrogen source	High-N fertilizer; rate and split timing must follow crop/local recommendation.
DAP	Nitrogen + phosphorus	Useful where P is required; do not treat as a universal fertilizer.
SSP	Phosphorus + sulphur	Can supply P and S; suitability depends on soil/crop requirement.
MOP / Potassium chloride	Potassium	Common K source; suitability depends on crop and soil conditions.
NPK complex fertilizers	N + P + K combinations	Use according to required nutrient ratio, not merely product name.
Zinc sulphate	Zinc	Use when Zn deficiency/low soil Zn is confirmed or locally recommended.
Gypsum	Calcium + sulphur	Important in selected crops/soils; groundnut is a common crop-specific context.
FYM / compost / vermicompost	Organic nutrient + organic matter	Supports integrated nutrient management; nutrient content is variable.
Green manure / crop residues	Organic nutrient cycling	Can improve soil organic matter and nutrient cycling.
Biofertilizers	Microbial nutrient support	Use crop- and organism-appropriate products and recommended application methods.

4. CROP-WISE NUTRIENT MANAGEMENT — RAG CARDS

Paddy / Rice

- Retrieve crop stage, establishment method, irrigation/water regime, soil-test information and local recommendation before giving an exact dose.
- Key nutrient themes: N for vegetative growth and yield formation; P for establishment/root development; K for plant strength, water regulation and stress response.
- Prefer split nutrient application where recommended rather than applying the entire N requirement at once.
- Yellowing questions should be cross-checked against N deficiency, Fe/Zn issues, waterlogging, root problems and disease.

Wheat

- Balanced N-P-K management is important; nitrogen timing should be aligned with crop growth stages and local recommendation.
- Consider soil-test results, previous crop and irrigation when interpreting fertilizer needs.
- Do not diagnose yellowing from leaf colour alone; compare pattern, plant age and field distribution.

Maize

- N is a major yield-related nutrient; P and K support establishment and crop development.
- Split N applications can improve nutrient-use efficiency where recommended.
- Watch for Zn deficiency in susceptible soils and confirm with soil/plant diagnosis before corrective application.

Groundnut

- Balanced nutrition supports vegetative growth and pod development.
- Calcium and sulphur are important crop-specific nutrient considerations, especially around pod development in suitable production systems.
- Gypsum should not be prescribed automatically; use crop/soil/local recommendation.

Cotton

- Balanced N, P and K are important; excessive N can cause excessive vegetative growth and affect crop balance.
- Micronutrient problems can occur in deficient soils; diagnose before applying corrective nutrients.
- Stage-specific nutrient management and local recommendations should guide exact rates.

Tomato

- Nutrition changes from establishment to vegetative growth, flowering and fruit development.
- N supports vegetative growth; P is important for roots and reproductive development; K is important for fruit development and quality.
- Calcium-related disorders should be distinguished from root/water problems and foliar diseases before corrective action.

Chilli

- Maintain balanced nutrition across vegetative, flowering and fruiting stages.
- Do not interpret leaf yellowing as a single nutrient deficiency without checking irrigation, root health and pests/diseases.
- Micronutrients should be applied only when deficiency/risk is supported by diagnosis or local recommendation.

Brinjal / Eggplant

- Balanced nutrition is needed throughout vegetative and fruiting stages.
- Excess nitrogen can produce excessive vegetative growth; exact nutrient rates should follow local recommendations.
- Combine nutrient diagnosis with pest/disease screening.

Onion

- Nutrient needs change with crop establishment and bulb development.
- Balanced N, P, K and sulphur management can be important depending on soil and production system.
- Avoid late excessive N where it can interfere with crop maturity/storage goals.

Potato

- Balanced N-P-K is important; K has a major role in crop quality and water relations.
- Exact fertilizer schedule depends on soil test, cultivar, season and production system.
- Distinguish nutrient deficiency symptoms from late blight and other foliar problems.

Sugarcane

- N, P and K are major nutrients; crop age and ratoon/plant crop status matter.
- Fertigation and split nutrient strategies may be relevant under suitable irrigation systems.
- Use soil/plant diagnostics and local recommendations for exact rates.

Banana

- Banana has substantial nutrient demand; requirements vary with plant age, cultivar and production system.
- N, K and other nutrients should be supplied according to crop stage and local recommendations.
- Drip/fertigation systems require schedule-specific recommendations rather than simply converting field doses.

Coconut

- Balanced nutrition supports palm growth and nut production; soil, palm age and yield level matter.
- Organic matter and site-specific nutrient management can complement mineral fertilizers.
- Use current local plantation recommendations for exact rates and timing.

Pulses

- Legume crops interact strongly with biological nitrogen fixation; nutrient management should account for crop and inoculation practices.
- P and S can be important in many pulse systems; micronutrients should be corrected when deficiency is established.
- Foliar nutrition should follow crop-specific recommendations rather than generic recipes.

5. NUTRIENT DEFICIENCY DIAGNOSIS

The assistant should treat visual symptoms as evidence, not proof. A good RAG response compares symptom pattern + crop stage + field distribution + soil/water conditions + pest/disease evidence.

Symptom / clue	Possible nutrient-related causes	Important differential checks
Older leaves yellow first	Often N deficiency; other mobile nutrient problems can also cause this.	Waterlogging, root damage, natural senescence, disease.
Young leaves yellow while veins remain green	Fe deficiency is a common diagnostic possibility.	High pH, root-zone problems, waterlogging, disease.
Interveinal chlorosis on older leaves	Mg deficiency can be considered.	K imbalance, disease, ageing and other stress.
Poor root/early growth	P deficiency may be considered.	Cold/wet soil, root disease, poor establishment.
Leaf margin scorching / necrosis	K deficiency may be considered.	Salinity, drought, chemical injury, disease.
Poor growing points / deformity	Ca or B deficiency can be considered depending on water stress.	Water stress, herbicide injury, pests and disease.
General yellowing	N deficiency is one possibility.	Never diagnose from colour alone.

6. FERTILIZER APPLICATION METHODS

- Basal application: nutrient supplied at or near planting/transplanting when recommended.
- Top dressing: fertilizer placed during crop growth, commonly used for split nutrient management.
- Band / localized placement: fertilizer positioned near the crop root zone while avoiding direct seed/root injury.
- Foliar application: nutrient solution applied to leaves; concentration and compatibility must be crop-specific.
- Fertigation: nutrients supplied through irrigation; concentration, frequency and compatibility must follow crop and irrigation recommendations.
- Seed treatment / biofertilizer inoculation: use only suitable products and recommended procedures.

7. EXACT DOSE SAFETY RULE FOR THE RAG

Never produce a universal exact fertilizer dose from memory. When a farmer asks "Urea evlo podanum?", first identify crop, location, season, variety/production system, soil-test result if available, irrigation method, crop stage and previous nutrient applications. If the retrieved official recommendation is location- and crop-specific, quote that recommendation with its conditions. Otherwise provide a safe general principle and ask for missing information.

Recommended response pattern: (1) identify likely nutrient need, (2) explain why, (3) state what information is missing for an exact dose, (4) retrieve/verify the current local recommendation, (5) mention application precautions.

8. RAG QUERY ROUTING

Farmer query	Retrieve
"Paddy-ku enna fertilizer?"	02 Crop Guide + 05 Fertilizer Guide + 01 Agriculture Guidelines
"Tomato leaves yellow ah irukku"	02 Crop Guide + 05 Fertilizer Guide + 03 Disease/Pest
"Groundnut-ku flowering time-la enna nutrient?"	02 Crop Guide + 05 Fertilizer Guide
"Urea evlo podanum?"	05 Fertilizer Guide + 01 Guidelines; ask crop/location/soil/stage
"Leaf yellow, fertilizer deficiency ah?"	05 Fertilizer + 03 Disease/Pest + 01 Guidelines
"Drip fertigation epdi?"	05 Fertilizer + 02 Crop Guide + 01 Guidelines
"Soil test report-la zinc low"	05 Fertilizer + 01 Guidelines; crop-specific correction
"Fertilizer subsidy / scheme irukka?"	04 Government Schemes + 05 Fertilizer Guide

9. RAG METADATA / CHUNKING TAGS

- document_type: fertilizer_nutrient_management
- topics: soil_testing, NPK, secondary_nutrients, micronutrients, fertilizer_types, INM, 4R, deficiency_diagnosis, fertigation, foliar_nutrition
- crops: rice, wheat, maize, groundnut, cotton, tomato, chilli, brinjal, onion, potato, sugarcane, banana, coconut, pulses

- location_priority: India; Tamil Nadu recommendations should be preferred when applicable
- answer_mode: explain + diagnose cautiously + retrieve local recommendation + verify exact dose
- freshness_rule: exact dose, product recommendation, subsidy, label direction or local schedule must be verified against current official guidance before presenting as current

10. OFFICIAL REFERENCE SOURCES

- ICAR — Balanced fertilizer use and soil-test-based nutrient management guidance (2026).
- ICAR — Soil-test-based balanced and integrated nutrient management; use of inorganic + organic nutrient sources.
- Tamil Nadu Agricultural University (TNAU) Agritech — crop and horticulture fertilizer schedules and plant nutrition resources.
- TNAU — research agenda covering integrated plant nutrient supply, nutrient deficiency management, fertigation and site-specific nutrient management.

Knowledge-base note: This PDF is intended for RAG retrieval and educational decision support. It is not a substitute for a current soil test, product label, local agricultural officer, or crop-specific official recommendation. Exact rates must be verified before field application.